Hao Li

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Research Interests

Analog and mixed signal circuit design; TIA and driver circuit design; DEM DAC circuit design; Analog computing architecture design; Low-pass filter circuit design based on VCO; Quantum computing circuit design based on VCO

Education

SUNY at Buffalo (advisor: Bibhu Datta Sahoo) Sep 2024 – Dec 2027

- **PhD in Electrical Engineering**|GPA: 4.00/4.00

Xi'an Jiaotong University, China Sep 2021 – May 2024

- **ME in Electrical Engineering**|GPA: 3.47/4.00

University of Electronic Science and Technology of China Sep 2017 – May 2021

- **BE in Electrical Engineering**|GPA: 3.78/4.00

Research Experience

SUNY at Buffalo – Research Assistant of NSF, Buffalo, NY Sep 2024 – Present

- Dynamic element matching DAC design in 65-nm CMOS TSMC, simulated in Cadence Virtuoso and Simulink
- Ring-VCO-based integrator design in 65-nm CMOS TSMC, which has been fabricated
- A 8×8 linear equation system based on VCO and memristor has been fabricated in 65-nm CMOS TSMC; this chip realizes 4×4 matrix inversion with 8-bit accuracy, achieving a $2us$ solution latency while consuming only $0.4mW$ power and $0.0146mm^2$ chip area from measurement results
- A 12-Qubit quantum computing chip based on VCO and memristor can run Grover's search algorithm and quantum fourier transform in 65-nm CMOS TSMC (simulated in Cadence Virtuoso, planning on tape-out)

Design of Silicon-based Optoelectronic Interface Driver Chip for Optical Computing, Xi'an, China Dec 2022 – May 2024

- A silicon-based optoelectronic interface driver chip with 1 Gb/s input rate fabricated in 180-nm CMOS TSMC, supporting up to 8-bit optoelectronic digital-to-analog conversion
- The chip mainly includes 8-channel Mach-Zehnder Modulator driver circuit and 8-segment MZM optical paths and has been fabricated

Pulsed TOF LiDAR Front-End Receiver Circuit Design, Xi'an, China Sep 2021 – Nov 2022

- The chip adopts a foldable stacked circuit structure within the channel to achieve current reuse and higher power efficiency in 180-nm CMOS TSMC and has been fabricated and now in commercial production
- It achieves full integration of low-power power supply and front-end circuit chips

Patent & Publication

Beyond MOSFETs: Overcoming Linearity Constraints in Analog Computing with Molecular FETs (submitted to IEEE Open Journal of Circuits and Systems) Oct 2026
Hao Li, Sai Sanjeet, M. Zaid Zaz, Peter Dowben, Jonathan Bird, and Bibhu Datta Sahoo

A $0.0146mm^2$, $0.4mW$ Ring Oscillator Based Analog Matrix-Inversion Macro for Massive MIMO in 65nm CMOS (submitted to JSSC) Oct 2026
Hao Li, Rizwan S. Peerla, Frank Barrows, Jonathan Bird, and Bibhu Datta Sahoo

Design of Analog SNNs with 4-Bit Programmable Synapses Using Molecular FETs (accepted by IEEE 69th MWSCAS Conference) Aug 2026
Sai Sanjeet, **Hao Li**, M. Zaid Zaz, Peter Dowben, Jonathan Bird, and Bibhu Datta Sahoo

A Ring Oscillator and Memristor Hybrid Architecture for High-Speed Analog Solution of a System of Linear Equations (accepted by IEEE Journal on Emerging and Selected Topics in Circuits and Systems) Jun 2026
Hao Li, Rizwan S. Peerla, Frank Barrows, Francesco Caravelli, and Bibhu Datta Sahoo

A T-DEM Based High-Speed Time-Interleaved Digital-to-Analog Converter (accepted by IEEE 68th MWSCAS Conference) Aug 2025
Hao Li and Bibhu Datta Sahoo

Comparison of Input Range and Area-Power Tradeoff of Fully Segmented and Highly Segmented DEM Digital-to-Analog Converters (accepted by IEEE 68th MWSCAS Conference) Aug 2025
Hao Li and Bibhu Datta Sahoo

A laser radar receiving front-end circuit, optical circuit and chip Mar 2025

- CN Patent App. CN202411636310.2A, where an effective solution has been implemented to reduce the overall power consumption caused by high current draw in the front-end receiver circuit

Multi-mode multi-line laser radar receiving circuit, receiving device and laser radar Aug 2024

- CN Patent App. CN202410664832.7A. The invention discloses a multimode multiline laser radar receiving circuit, a receiving device, and a laser radar, which belong to the technical field of integrated circuits

A 250-kHz-BW 15.8-ENOB 2nd-Order Noise-Shaping SAR ADC With Multi-Path-Input Floating Inverter Amplifier Oct 2023

- Conference paper, the SAR ADC achieves 97.15-dB SNDR and consumes only 3.76 mW under the 3.3-V power supply, which leads to an Schreier FoMs of 175.4 dB

Optical Receiver Front-End for 50G PON in 40nm CMOS Oct 2022

- Conference paper, we design a 50Gb/s transimpedance amplifier (TIA) circuit with very low power consumption, which greatly reduces the manufacturing cost by adopting the CMOS process

A Short Circuit Current Limiting and Protection Circuit for High Side Switches Sep 2022

- Journal paper, this paper presents a short-circuit current limiting and protection circuit for high side switch, the mode of secondary protection was adopted by the circuit

IToF and DToF dual-mode VCSEL driving circuit, module, chip, method and equipment Aug 2022

- CN Patent CN114976858B, which discloses IToF and DToF dual-mode VCSEL driving circuits, modules, chips, methods and equipment

Awards

- 2026 IEEE Circuits and Systems Student Travel Grant
- Student Poster Award, 3rd Place in Workshop on Emerging Trends in Semiconductor Technology
- Outstanding Graduate Student of Xi'an Jiaotong University
- National Endeavor Fellowship
- Outstanding Student Scholarship of University of Electronic Science and Technology of China

Reviewers

- IEEE TCAS-I x 3
- IEEE OJCAS x 2
- IEEE ISCAS x 4
- IEEE MWSCAS x 8
- IEEE VLSID x 3

Languages

- English (Fluent)
- Mandarin Chinese (Native)